

Progress Report

Depth of Field Research

Rui Gao — Sep 27

Week 5 · Sep 21 – Sep 27, 2026

Progress

- Researched traditional DoF techniques: gather and scatter filters, layered DoF, multi-view rendering, and lens-sampled ray tracing.
- Surveyed learning-based work: neural bokeh rendering (BokehMe, Dr.Bokeh) and monocular depth estimation (MiDaS, Depth Anything).
- Built a procedural café scene with 126 objects, shadows, interactive focus and aperture controls, and DoF, depth, and CoC views.
- Compared three DoF methods on one scene against a ray-traced reference: the screen-space gather has 4.95/255 mean error versus 0.68/255 for multi-view, and more samples barely help the gather.
- Started physically based rendering with Mitsuba 3 (to run on the HPC): first renders of a realistic café with global illumination, soft shadows and glass, plus ground-truth DoF and depth.

Meeting — Sep 21

Attendees: Rui Gao, Beibarys Myrzash, Professor Lingqi Yan.

- Presented the simple CoC-based DoF approach and its 3D scene; discussed preparing for dataset generation.
- Received next tasks: build a better, more realistic scene; learn a physically based renderer; and define the dataset's shape and attributes (image resolution, depth maps, focus and aperture settings, camera metadata).

Next Steps

- Run large, high-quality renders on the HPC.
- Plan a dataset of sharp images, depth maps, and ground-truth DoF across focus and aperture settings.
- Test monocular depth estimation as the depth source for DoF.